

Article

Comparative Analysis and Noise Robustness Study of Quantum Kernel Methods and Variational Quantum Classifiers for Financial Fraud Detection

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Abstract

Quantum machine learning on near-term noisy quantum devices has generated substantial theoretical interest, but rigorous empirical comparisons under realistic noise on practically relevant data remain scarce. This paper compares two paradigmatic quantum learning models, a Quantum Support Vector Machine (QSVM) built on the ZZFeatureMap quantum kernel and a Variational Quantum Classifier (VQC) with an EfficientSU2/RealAmplitudes ansatz, against tuned classical baselines (SVM with four kernels, Random Forest, XGBoost, LightGBM and CatBoost) on the ULB Credit Card Fraud dataset (284,807 transactions, 0.17% fraud). All models share an identical 4-qubit PCA-reduced feature space, evaluated on the full unbalanced test fold over 15 fits (3 folds $\times$ 5 seeds) and reported as mean $\pm$ standard deviation with bootstrap confidence intervals, AUPRC as the primary metric. Noise robustness is assessed under depolarizing noise $p \in \{0, 0.001, 0.01, 0.05\}$, with ranking preservation measured directly through Spearman ρ and Kendall τ between the noisy and noiseless decision scores rather than read off AUPRC, alongside the per-paradigm computational cost. At four qubits the classical baselines lead (AUPRC 0.60 to 0.74, CatBoost best), above the VQC (0.494) and the QSVM (0.240); the controlled QSVM-versus-RBF-SVM comparison puts the cost of the quantum kernel at about 0.45 AUPRC. Under noise the QSVM keeps its score ranking ($\rho = 0.998$ at $p = 0.001$, 0.906 at $p = 0.01$) and an operational decision threshold (recall 0.87 to 0.89, stable calibration), while the VQC AUPRC peaks non-monotonically at $p = 0.01$ (0.494 rising to 0.654, then easing to 0.569 at $p = 0.05$) even as its ranking decays monotonically (ρ from 0.72 to near zero), so average precision on its own misrepresents how noise affects it. The quantum models do not surpass the tuned classical reference at four qubits; the contribution is methodological: under noise, AUPRC has to be read together with a genuine rank statistic, because the two can move in opposite directions.



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Keywords: quantum machine learning; quantum kernel methods; variational quantum classifier; QSVM; VQC; fraud detection; noise robustness; NISQ; depolarizing noise; computational cost

1. Introduction

Financial fraud detection is one of the most practically important and technically challenging classification problems in machine learning. The Credit Card Fraud dataset published by the Université Libre de Bruxelles (ULB) [1] contains 284,807 transactions of

which only 492 are fraudulent, yielding a class imbalance ratio of approximately 577:1. This severe imbalance, combined with the non-stationary nature of fraud patterns and strict operational latency requirements, makes the domain a demanding test bed for new learning paradigms [2,3].

The advent of NISQ devices [4] has stimulated significant research into QML [5,6]. Two paradigms have emerged as leading candidates for near-term quantum classification: quantum kernel methods, where a quantum circuit defines a feature map into an exponentially large Hilbert space and classical SVM optimization finds the separating hyperplane [7,8]; and variational quantum classifiers (VQCs), which employ parameterized quantum circuits trained via hybrid quantum–classical optimization loops [9–11].

Despite substantial theoretical interest, empirical evidence for quantum advantage in practical classification tasks remains limited. Recent theoretical analyses [12–14] suggest that quantum speedup in supervised learning is highly problem-dependent and potentially restricted to specially structured datasets. The phenomenon of exponential concentration of quantum kernels, wherein kernel values become uniformly near-zero as the number of qubits grows, further complicates the scaling of quantum kernel methods [15]. On the hardware side, noise inherent to current NISQ devices poses a fundamental barrier to practical deployment [16,17].

This work contributes to the empirical benchmarking literature through a controlled, reproducible study on the full ULB Credit Card Fraud dataset. Our primary contributions are:

1. A two-level comparison framework: a structural comparison between QSVM and its classical SVM RBF counterpart (isolating the effect of the quantum kernel), and a broader comparison against the best classical baselines (SVM Linear, Random Forest, XGBoost, LightGBM, CatBoost).
2. An extended hyperparameter study covering 10 VQC configurations and a QSVM regularization sweep, all evaluated with AUPRC as primary metric on the full, unsampled test set.
3. A noise robustness analysis under depolarizing noise at levels $p \in \{0, 0.001, 0.01, 0.05\}$, using density matrix simulation, revealing a qualitative asymmetry in degradation behavior between QSVM and VQC.
4. A computational cost analysis quantifying training time, inference time, circuit depth, and scaling across $N \in \{4, 6, 8, 10\}$ qubits.

The paper is structured as follows: Section 2 reviews the theoretical background. Section 3 describes the experimental pipeline and design decisions. Section 4 presents the results. Section 5 interprets the findings and discusses limitations, and Section 6 concludes.

2. Background and Related Work

2.1. Quantum Computing Fundamentals

A qubit is the basic unit of quantum information, existing as a superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where $|\alpha|^2 + |\beta|^2 = 1$ [18]. An n -qubit system inhabits a 2^n -dimensional complex Hilbert space. Quantum gates are unitary transformations; key single-qubit gates include the Hadamard gate H and Pauli rotations R_X, R_Y, R_Z , while the two-qubit CNOT gate generates entanglement, producing correlations with no classical analog [18].

2.2. Quantum Kernel Methods and QSVM

Quantum kernel methods [7,8] embed classical data $\mathbf{x} \in \mathbb{R}^n$ into a quantum feature space via a parameterized circuit $U_\phi(\mathbf{x})$, defining the quantum kernel:

$$k(\mathbf{x}_i, \mathbf{x}_j) = |\langle 0^n | U_\phi^\dagger(\mathbf{x}_j) U_\phi(\mathbf{x}_i) | 0^n \rangle|^2 \quad (1)$$

This fidelity-based kernel corresponds to the transition probability between two quantum states. The ZZFeatureMap [7] encodes data via second-order Pauli Z-Z interactions:

$$U_{\phi}(\mathbf{x}) = \exp\left(i \sum_{jk} \phi_{jk}(\mathbf{x}) Z_j Z_k\right) \exp\left(i \sum_j x_j Z_j\right) \quad (2)$$

where $\phi_{jk}(\mathbf{x}) = (\pi - x_j)(\pi - x_k)$ captures pairwise feature correlations. The kernel matrix is then passed to a classical SVM via `sklearn.svm.SVC(kernel='precomputed')` [19,20]. A known failure mode is exponential concentration: for deep circuits or large n , kernel values concentrate near $1/2^n$, rendering the kernel uninformative [15].

2.3. Variational Quantum Classifiers

VQCs [9–11] employ a parameterized quantum circuit $U(\theta)$ combining a feature map $U_{\phi}(\mathbf{x})$ and a trainable ansatz $W(\theta)$:

$$f(\mathbf{x}; \theta) = \langle 0^n | U_{\phi}^{\dagger}(\mathbf{x}) W^{\dagger}(\theta) \hat{O} W(\theta) U_{\phi}(\mathbf{x}) | 0^n \rangle \quad (3)$$

Parameters θ are updated classically to minimize cross-entropy loss. Standard ansatzes include RealAmplitudes (real-valued, hardware-efficient) [21] and EfficientSU2 (arbitrary SU(2) rotations with full entanglement) [22]. Training uses gradient-free optimizers, specifically COBYLA [23] or SPSA [24], to avoid barren plateau issues [25].

2.4. Noise Models in NISQ Devices

NISQ devices suffer from gate errors, decoherence, and readout errors. The depolarizing channel models gate-level noise as:

$$\mathcal{E}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \quad (4)$$

where p is the error probability per gate [18]. Density matrix simulation, as opposed to statevector simulation, is required to accurately model mixed quantum states under noise. VQC robustness under depolarizing noise is analyzed theoretically in [26]; quantum kernel degradation under noise is studied in [17]. Current superconducting qubit processors report single-qubit gate errors of 10^{-3} to 10^{-4} and two-qubit gate errors of 10^{-2} to 10^{-3} [27], which directly motivates our choice of noise levels.

2.5. Classical Baselines

SVM [19,28] finds the maximum-margin hyperplane in a kernel-induced feature space and remains a workhorse for applied classification and monitoring problems, from robust multivariate process monitoring with probability-of-control schemes [29] to comparative risk studies on photovoltaic-system control [30]. The RBF kernel is the direct classical counterpart to the quantum fidelity kernel, making the QSVM vs. SVM RBF comparison the central structural test of this study. Random Forest [31] ensembles decorrelated decision trees via bagging. Gradient-boosted decision trees are represented by XGBoost [32], LightGBM [33], and CatBoost [34]; the three differ in how they grow trees (depth-wise versus leaf-wise) and how they handle categorical and high-cardinality features, but reach near-ceiling performance on tabular fraud-detection benchmarks [3].

2.6. Related Work

The ULB Credit Card Fraud dataset [1,2] is the standard benchmark for financial anomaly detection. AUPRC is the standard primary metric for imbalanced settings, as it focuses on the minority class and is insensitive to the large number of true negatives that

inflate AUROC [2]. Quantum approaches to finance [35,36] have focused primarily on portfolio optimization; systematic reviews of the QML landscape [36] and AI-assisted bibliometric analyses of the adjacent scientific literature [37] document its rapid expansion. Kyriienko et al. [38] applied QSVM to small financial datasets without noise evaluation. Bowles et al. [39] survey QML benchmarking pitfalls, reinforcing the need for controlled comparisons such as the one presented here.

Supervised quantum machine learning for fraud detection is a small but rapidly growing area. The work most directly comparable to ours is El Alami et al. [40], the only prior study to use the same ULB European credit-card dataset (284,807 transactions, 0.172% fraud) with PCA to seven features and VQC/QNN models at four to seven qubits. We differ on four methodological axes. (i) Class balance: El Alami et al. balance the classes before training and report a single best $F_1 = 0.88$, whereas we preserve the native $\approx 577:1$ imbalance and adopt AUPRC as the primary metric, which is the deployment-relevant regime. (ii) Statistical reporting: they report point estimates on a single split, whereas we report 15 fits (3×5) across more than 30 metrics with 95% bootstrap confidence intervals. (iii) Classical baselines: they do not benchmark against strong classical models, whereas we tune SVM (four kernels), Random Forest, XGBoost, LightGBM and CatBoost. (iv) Noise robustness: they do not characterize noise, whereas we sweep depolarising noise $p \in \{0, 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$ with paired ranking and decision metrics. The two studies are therefore complementary rather than duplicative.

Two further points position our findings. Chaves et al. [41] report average precision 0.793 ± 0.085 for an autoencoder-VQC mixture-of-experts versus 0.770 ± 0.096 for an XGBoost baseline on a comparable imbalanced dataset, suggesting that quantum components help only when embedded in a feature-engineered hybrid, consistent with our finding that raw QSVM/VQC at four qubits do not surpass tuned gradient boosting. Finally, Kyriienko and Magnusson [38] report a 15% average-precision separation in favor of a re-uploading quantum kernel, but at 20 qubits; no advantage is claimed at the small qubit counts we study, which directly supports our negative result. Throughout, we follow the benchmarking discipline of Bowles et al. [39], who show that single-seed QML comparisons are unreliable and that classical baselines win on average across diverse datasets.

3. Methodology

3.1. Dataset and Preprocessing Pipeline

The Credit Card Fraud Detection dataset [1] contains 284,807 transactions of which 492 (0.173%) are fraudulent. Features V1 through V28 are the result of a PCA transformation applied by the original data providers to protect cardholder identity; Amount and Time are the raw original features. The severe class imbalance ($\approx 577:1$) means that a naive classifier predicting all samples as legitimate achieves 99.83% accuracy, making accuracy a misleading metric and motivating the use of AUPRC throughout.

Figure 1 illustrates the class distribution. The preprocessing pipeline was designed to prevent data leakage and ensure fair comparison across all models.

The pipeline proceeds in strict sequential order. First, StandardScaler is applied to Amount and Time; V1–V28 are already zero-centered from the original PCA. Second, the data are partitioned by 3-fold stratified cross-validation applied to the full population before any subsampling, so that each held-out test fold ($\approx 94,936$ transactions) preserves the original class distribution; the full 15-fit protocol crossing these folds with 5 seeds is detailed in Section 3.7 (Statistical Protocol). Partitioning before subsampling is critical: subsetting the test fold would yield artificially inflated metrics, as the model would never encounter the realistic 0.17% fraud rate at inference time. Third, random undersampling of the majority class is applied to the training set only, targeting a 1:1 fraud-to-legitimate ratio.

The 1:1 ratio was chosen over 1:3 because quantum kernel computation scales as $O(m^2)$ in the number of training samples; with a 1:3 ratio the kernel matrix computation would require approximately 9 times more evaluations, making the experiments computationally infeasible within reasonable time budgets on the simulator.

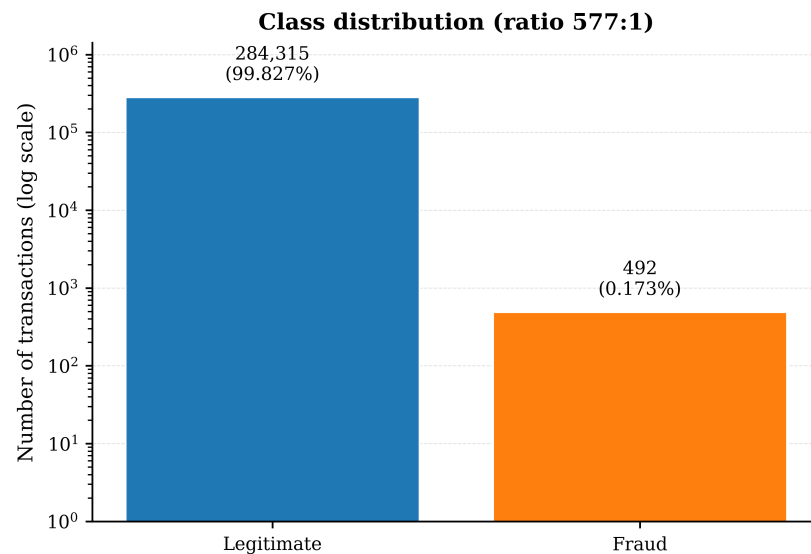


Figure 1. Class distribution in the Credit Card Fraud dataset. The ratio of 577:1 (legitimate to fraudulent) makes accuracy a misleading metric; AUPRC is used throughout as the primary evaluation criterion.

Fourth, PCA is applied to reduce from 30 features to N components. Figure 2 shows that $N = 4$ retains 89.1% of explained variance, $N = 6$ retains 93.3%, $N = 8$ retains 94.9%, and $N = 10$ retains 96.0%. Fifth, for quantum circuits only, an additional MinMaxScaler is applied to map features to $[0, \pi]$. This step is specific to quantum encoding and serves two purposes: it prevents aliasing in qubit rotation angles (values outside $[0, \pi]$ would be equivalent modulo 2π to smaller values, losing information), and it mitigates exponential concentration of quantum kernels by keeping inner products between encoded states informative [15].

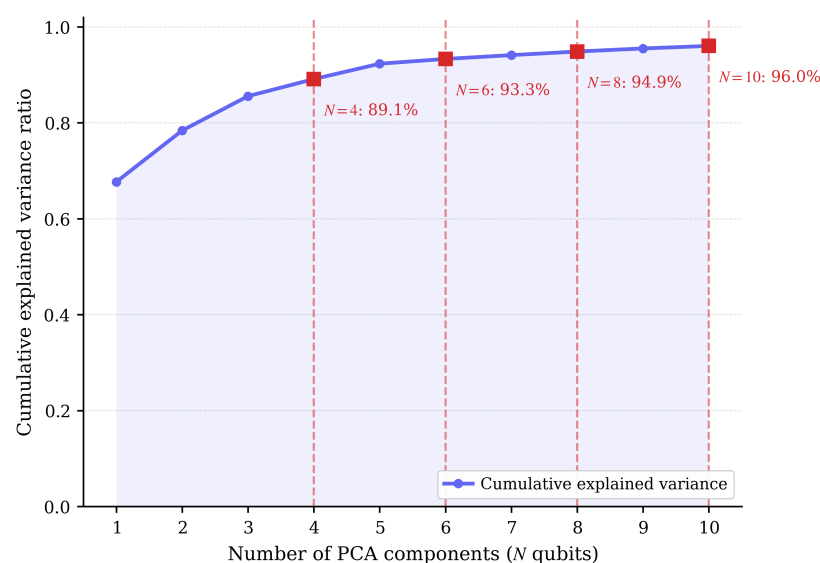


Figure 2. PCA explained variance as a function of number of components. At $N = 4$ components (4 qubits), 89.1% of variance is retained. The marginal gain per additional component decreases monotonically, supporting the choice of $N = 4$ as a computationally feasible operating point.

3.2. Choice of $N = 4$ Qubits

The four-qubit operating point emerged from a sweep across $N \in \{4, 6, 8, 10\}$. Figure 3 shows the PCA scatter plots at each dimensionality: the two classes remain reasonably separable at $N = 4$, validating that the dimensional reduction does not eliminate discriminative structure.

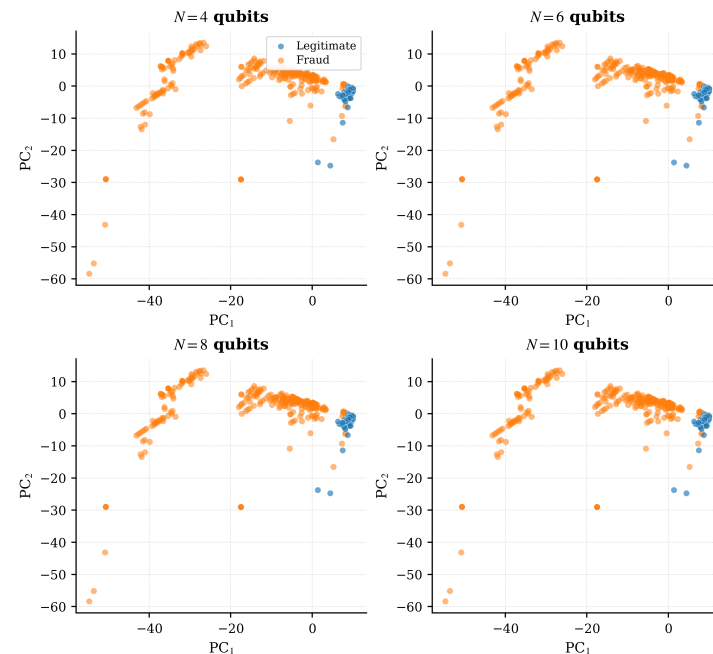


Figure 3. PCA scatter plots for $N \in \{4, 6, 8, 10\}$ qubit counts. Even at $N = 4$, the fraudulent class (orange) occupies a distinct region in the principal component space, confirming that sufficient discriminative information is retained.

Classical models (SVM RBF and Random Forest) showed marginal AUPRC differences (± 0.02) across all four qubit counts, confirming that the dimensional reduction is not the performance-limiting factor. For quantum models, preliminary kernel matrix computations at $N \geq 8$ revealed near-degenerate matrices with off-diagonal values approaching $1/2^N$, a direct manifestation of exponential concentration [15]. At $N = 4$, as shown by the bimodal kernel value distribution in Figure 4, the kernel retains meaningful inter-class discrimination. This makes $N = 4$ the natural operating point: the smallest qubit count at which the quantum kernel remains informative.

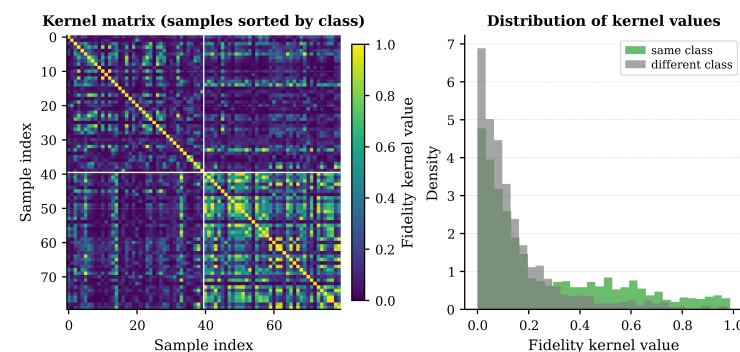


Figure 4. Quantum kernel matrix for QSVM at 4 qubits ((Left): samples sorted by class; (Right): kernel value distribution). The bimodal distribution of fidelity values, with intra-class pairs (same label) concentrating at higher values than cross-class pairs, confirms that the ZZFeatureMap kernel encodes class-relevant geometric structure at $N = 4$. This structure disappears at $N \geq 8$ due to exponential concentration.

3.3. QSVM Implementation

The QSVM uses a fidelity quantum kernel with the ZZFeatureMap (reps = 2, full entanglement pattern) on 4 qubits, implemented with Qiskit [22]. How the kernel is evaluated, from exact statevectors at $p = 0$ and from density matrices at $p > 0$, is set out in Section 3.6; Figure 5 shows both circuits (panel a is the encoding map). The architecture was fixed from an initial single-split ablation: reps = 2 improved average precision over reps = 1, while reps = 3 started to show concentration effects with no further gain. A regularization sweep over $C \in \{1, 10, 50, 100, 500, 1000\}$ selected $C = 500$, well above the default $C = 1$, which reflects the tighter SVM margin needed for the imbalanced feature distribution. The kernel matrix is computed for the train–train pair (for fitting) and the test–train pair (for inference) and passed to `sklearn.svm.SVC(kernel='precomputed')`. Average-precision results under this fixed architecture, over the full multi-seed protocol, are reported in Section 4.

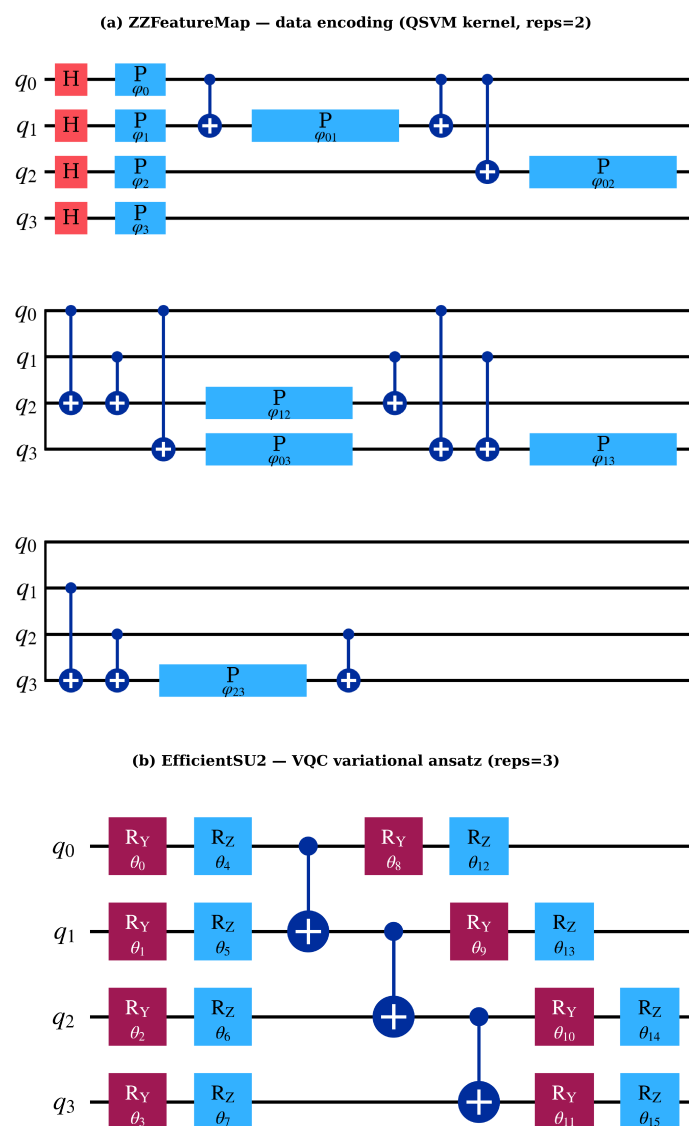


Figure 5. The two quantum architectures. (a) The ZZFeatureMap data-encoding circuit: a Hadamard layer, single-qubit phase gates $P(\varphi_i)$ with $\varphi_i = 2x_i$, and pairwise ZZ entangling phases $P(\varphi_{ij})$ with $\varphi_{ij} = 2(\pi - x_i)(\pi - x_j)$; the QSVM fidelity kernel uses this map at reps = 2. (b) The EfficientSU2 variational ansatz: per-qubit $R_Y(\theta_j)$ and $R_Z(\theta_j)$ rotations with trainable parameters θ_j and a linear CX entangling ladder; the VQC uses reps = 3. Both circuits are drawn at reps = 1 for legibility, and panel (a) is folded into rows to keep its gate columns readable; the symbolic phase labels φ_i and φ_{ij} (panel (a)) and the trainable angles θ_j (panel (b)) are printed at each gate.

3.4. VQC Implementation

The VQC uses the same ZZFeatureMap as the QSVM for data encoding, ensuring a fair comparison of the classification head rather than the encoding. After the feature map, a trainable ansatz $W(\theta)$ is appended; the full circuit implements $W(\theta) \cdot U_\phi(\mathbf{x})|0^n\rangle$ and the classification decision is based on measuring the expectation value of the Z observable on the first qubit.

We evaluated 7 standard VQC configurations and 3 data re-uploading variants [42]. The standard configurations combine two ansatz types (RealAmplitudes and EfficientSU2) with depths $reps \in \{2, 3\}$ and optimizers COBYLA and SPSA, with up to 500 optimization iterations. Figure 6 summarizes the AUPRC results across configurations.

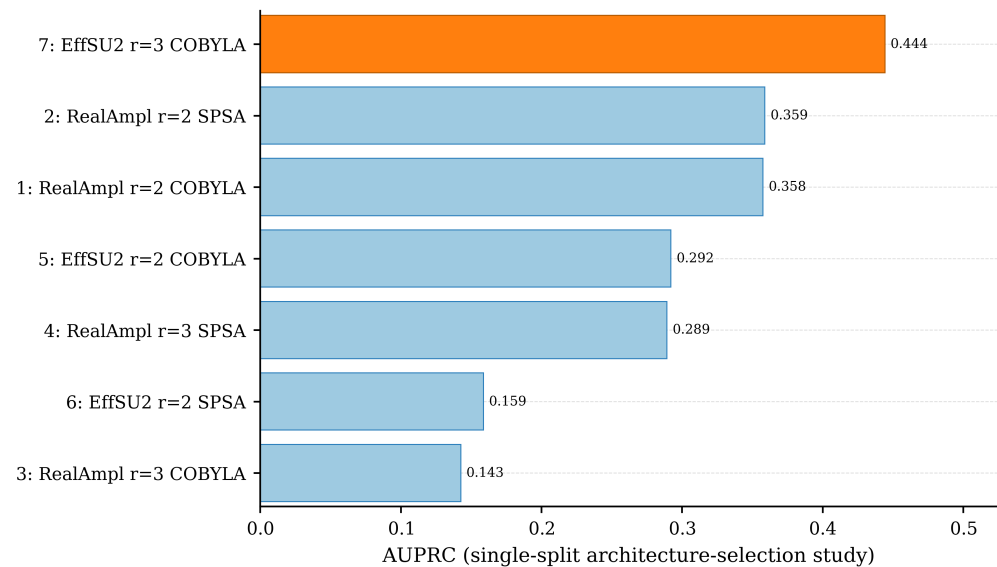


Figure 6. VQC hyperparameter comparison at four qubits across the standard ansatz/optimizer configurations, from a single-split architecture-selection study. EfficientSU2 with $reps = 3$ and the COBYLA optimiser (highlighted) gives the highest average precision, and this is the architecture carried forward to the multi-seed evaluation in Section 4, whose reported numbers use the full protocol. Three data re-uploading variants were also evaluated but are omitted here; they suffered from optimisation instability with final loss values around 0.81.

The best configuration, EfficientSU2 with $reps = 3$ and COBYLA, contains 32 trainable parameters. EfficientSU2 outperforms RealAmplitudes because it allows full SU(2) rotations per qubit (two rotation angles R_Y and R_Z per qubit per layer), providing greater expressivity. COBYLA outperforms SPSA for this problem because COBYLA constructs a local quadratic model of the objective, which converges faster for the relatively low-dimensional parameter space (≤ 32 parameters). SPSA would be preferable for larger circuits where analytical or finite-difference gradients become expensive, but at 32 parameters COBYLA's model-based approach is more sample-efficient. The EfficientSU2 ansatz is panel (b) of Figure 5, and Figure 7 shows the training loss convergence.

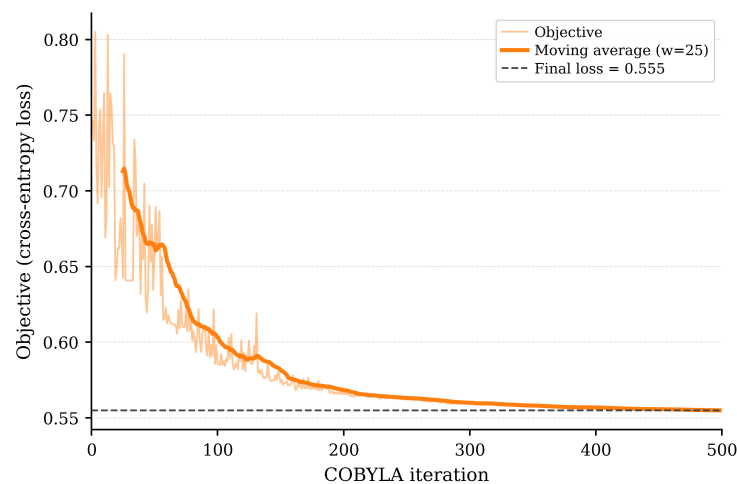


Figure 7. Training loss convergence for the best VQC (EfficientSU2, reps = 3, COBYLA, 32 parameters). The optimizer converges to a final loss of 0.555 after approximately 400 iterations, with no oscillations suggesting a clean convergence basin. The gap between 0.555 and the theoretical minimum of 0 reflects the inherent difficulty of the fraud detection task at 4 qubits.

3.5. Noise Robustness Protocol

To assess NISQ-era practicality, depolarizing noise is introduced via Qiskit’s Noise-Model applied to AerSimulator in density matrix mode. Density matrix simulation is required rather than statevector simulation because noise introduces mixedness: the quantum state is no longer a pure state and must be represented as a $2^n \times 2^n$ matrix rather than a 2^n amplitude vector [18]. The three noise levels tested, $p \in \{0.001, 0.01, 0.05\}$, were chosen to span the range from current best hardware error rates ($\sim 10^{-3}$ for single-qubit gates [27]) to error rates representative of early NISQ devices without error mitigation.

For QSVM, the quantum kernel matrix is fully recomputed under each noise model. For VQC, inference is performed with noisy circuits using weights trained under noiseless conditions, following the protocol in [26]. Noise robustness is evaluated on the same full unbalanced test fold ($\approx 94,936$ transactions) used throughout, which the kernel implementation of Section 3.6 keeps tractable under density-matrix simulation.

3.6. Quantum Kernel Implementation and Comparability

The QSVM kernel is evaluated through two complementary classical-simulation paths, dispatched by the noise level. In the noiseless regime ($p = 0$) we use a fidelity kernel computed from exact statevectors, $K_{ij} = |\langle \phi(x_i) | \phi(x_j) \rangle|^2$, following the $O(N)$ statevector construction of Kyriienko and Magnusson [38]; this replaces the $O(N^2)$ circuit-pair construction of the naive FidelityQuantumKernel and makes the full $n_{\text{test}} = 94,936$ evaluation tractable. For $p > 0$ the kernel is computed from density matrices, $K_{ij} = \text{Re Tr}(\rho_i \rho_j)$, propagated through the depolarizing channel with an AerSimulator in density_matrix mode, following Wang et al. [43]. The two paths coincide at $p = 0$ to within $\text{atol} = 10^{-10}$ (verified by a unit test) and join continuously as $p \rightarrow 0$, so the noise sweep is internally consistent.

Per-fit wall-clock cost (single-thread, $n_{\text{train}} = 656$) reflects the two paradigms’ distinct computational structures: QSVM requires ≈ 3.8 min at $p = 0$ (statevector kernel) and ≈ 25.1 min at $p > 0$ (density-matrix kernel), whereas VQC requires ≈ 14.2 min at $p = 0$ and ≈ 10.5 min at $p > 0$ (COBYLA, $\text{max_iter} = 500$). We stress that these figures are not an apples-to-apples runtime comparison: QSVM is a kernel method, a single convex support-vector solve on a precomputed Gram matrix [44], while VQC is an iterative variational optimization. Each paradigm is implemented with its own best-practice classical-simulation strategy, and we compare the two on predictive metrics, not on runtime.

Three methodological asymmetries are worth flagging, each a deliberate per-paradigm choice:

1. Hyperparameter selection. Each classical baseline is reported at its best configuration over a hyperparameter sweep (e.g., CatBoost reaches AUPRC = 0.741 at $n_{\text{est}} = 100$), whereas the quantum models use fixed architectures (ZZFeatureMap reps = 2; EfficientSU2 reps = 3). The boosted models in fact peak at the low end of the n_{est} sweep, an overfitting signature we note in Section 4.1; the classical baseline is therefore a strong, well-tuned reference.
2. Sampling noise. The VQC inference path draws 1024 shots from the noisy sampler (finite-sampling realism for a variational classifier), while the QSVM noisy kernel is integrated exactly over the density matrix (shot-free). Part of the faster ranking degradation we observe for VQC at $p \geq 0.01$ thus reflects shot noise compounding the depolarizing channel, and we report both contributions.
3. Kernel dispatch. The two-path kernel above (statevector for $p = 0$, density-matrix for $p > 0$) is the QSVM analogue of the VQC's fixed sampler: each is the standard, best-available simulation route for its regime.

3.7. Statistical Protocol

To ensure statistical robustness, every (model, noise level) condition is evaluated over $N_{\text{fits}} = 15$ independent fits, obtained from a 3-fold stratified cross-validation (applied to the full population before any undersampling) crossed with 5 independent random seeds per fold. The effective seed $s_{\text{eff}} = 1000k + s$ (fold index k , seed s) controls all stochastic components jointly: the random undersampling of the majority class, the initialization of the VQC variational parameters, the stochastic steps of the COBYLA optimiser, and the random state of the classical baselines. Each test fold preserves the native class distribution ($\pi \approx 0.17\%$ fraud) and contains $\approx 94,936$ transactions (of which ≈ 164 are fraudulent), so no information about the imbalance is discarded at evaluation time.

For each fit we compute a panel of more than 30 metrics organized in nine tiers: ranking (AUPRC, AUROC, partial AUROC at $\text{FPR} \leq 0.01$), decision quality at the 0.5 threshold and at the F_1 -optimal threshold (precision, recall, F_1 , F_2), imbalance-robust scores (Matthews correlation coefficient [45], balanced accuracy, geometric mean), calibration (Brier score, expected calibration error [46], log loss), top- K operational metrics (precision among the 100 and 1000 highest-ranked transactions), and, for the noisy settings, ranking-preservation statistics (Spearman ρ and Kendall τ between the noisy and noiseless decision scores). Results are reported as mean $\pm$ standard deviation over the 15 fits, complemented by 95% confidence intervals obtained from a stratified percentile bootstrap (1000 resamples of the per-fit predictions, preserving class prevalence in every resample [47]).

This $3 \times 5 = 15$ -fit design follows the repeated-cross-validation protocol of Chaves et al. [41], who report average precision as 0.793 ± 0.085 over the same scheme on a comparable imbalanced credit-card dataset. More broadly, reporting dispersion across seeds and multiple complementary metrics, rather than a single point estimate, follows the benchmarking discipline urged by Bowles et al. [39], whose study documents accuracy swings exceeding 14% from minor design choices and warns explicitly against single-seed reporting.

4. Experimental Results

4.1. Classical Baseline Results

Table 1 reports the four SVM kernels on the 4-qubit PCA features. The linear kernel, not the RBF, gives the highest average precision (0.720). That fits the feature space: V1–V28 are already PCA components produced by the data providers, so they are decorrelated and roughly Gaussian, and a linear boundary is easier to place there than in a raw feature space, while the RBF kernel’s infinite-dimensional lifting can overfit. This also sets up the QSVM comparison later: when a feature space is already well separated by a linear boundary, an extra nonlinear quantum-kernel embedding on top has limited room to add useful structure, which is what the QSVM-versus-RBF result in Section 4 confirms.

The low precision across all four kernels (0.10–0.35) reflects the imbalance: even at the best threshold, a 577:1 ratio means a sensitive classifier raises many false alarms. The polynomial kernel is an instructive outlier: it has the highest F1 (0.478), precision (0.354) and Matthews correlation (0.524) of the four, yet the lowest AUPRC (0.597, last row of Table 1). A better-placed threshold does not imply better ranking, which is why we read AUPRC first and threshold-dependent scores second.

Table 1. Support vector machine baselines on the four-qubit PCA features (linear, RBF, polynomial, and sigmoid kernels), evaluated noise-free at $n_{\text{train}} = 656$ on the full unbalanced test fold. Values are mean $\pm$ standard deviation over 15 fits (3 splits $\times$ 5 seeds). The linear kernel gives the highest average precision of the four.

Model	AUPRC	AUROC	Precision	Recall	F1	MCC
SVM (linear)	0.720 $\pm$ 0.019	0.971 $\pm$ 0.016	0.163 $\pm$ 0.182	0.862 $\pm$ 0.043	0.245 $\pm$ 0.168	0.345 $\pm$ 0.140
SVM (sigmoid)	0.699 $\pm$ 0.061	0.970 $\pm$ 0.015	0.106 $\pm$ 0.107	0.864 $\pm$ 0.038	0.175 $\pm$ 0.129	0.281 $\pm$ 0.109
SVM (RBF)	0.690 $\pm$ 0.077	0.972 $\pm$ 0.014	0.101 $\pm$ 0.092	0.869 $\pm$ 0.041	0.169 $\pm$ 0.123	0.274 $\pm$ 0.103
SVM (polynomial)	0.597 $\pm$ 0.035	0.964 $\pm$ 0.023	0.354 $\pm$ 0.141	0.813 $\pm$ 0.015	0.478 $\pm$ 0.147	0.524 $\pm$ 0.118

We extend the baseline suite with LightGBM [33] and CatBoost [34], two strong gradient-boosting libraries, each swept over $n_{\text{est}} \in \{100, 200, 500, 1000, 2000\}$. At $n_{\text{train}} = 656$ and zero noise, CatBoost attains the highest mean average precision in our sweep, $\text{AUPRC} = 0.741 \pm 0.043$, followed by LightGBM (0.708 ± 0.047) and XGBoost (0.706 ± 0.051); Random Forest trails at 0.679 ± 0.080 (Table 2). We deliberately describe CatBoost as the best model in our sweep rather than state of the art: the absolute values remain dataset- and split-dependent, and the gap to LightGBM and XGBoost is well within one standard deviation. The three boosting models reach their best AUPRC at the low end of the n_{est} range ($n_{\text{est}} = 100$ for LightGBM and CatBoost, 200 for XGBoost), with performance falling as more trees are added (an overfitting signature on this imbalanced, PCA-reduced feature space), whereas Random Forest improves monotonically up to $n_{\text{est}} = 2000$. These tuned gradient-boosting baselines (AUPRC 0.68–0.74) set a strong classical reference against which the quantum models are measured in Section 5.4; both quantum paradigms fall below this band at four qubits, a result we take up in the Section 5.

Table 2. Gradient-boosted and Random Forest baselines across the number-of-trees sweep, noise-free at $n_{\text{train}} = 656$; n_{est} is the number of estimators. Values are mean $\pm$ standard deviation over 15 fits. For the boosted models the best average precision sits at the low end of the sweep, so adding trees does not help here and slightly hurts.

Model	Hyperparameters	AUPRC	AUROC	F1	MCC	Brier
CatBoost	$n_{\text{est}} = 100$	0.741 ± 0.043	0.977 ± 0.012	0.088 ± 0.015	0.199 ± 0.017	0.031 ± 0.005
CatBoost	$n_{\text{est}} = 200$	0.724 ± 0.043	0.978 ± 0.010	0.073 ± 0.014	0.180 ± 0.014	0.033 ± 0.007
CatBoost	$n_{\text{est}} = 500$	0.707 ± 0.048	0.976 ± 0.011	0.067 ± 0.013	0.171 ± 0.013	0.035 ± 0.008
CatBoost	$n_{\text{est}} = 1000$	0.691 ± 0.046	0.976 ± 0.011	0.064 ± 0.013	0.167 ± 0.013	0.038 ± 0.008
CatBoost	$n_{\text{est}} = 1500$	0.688 ± 0.046	0.975 ± 0.011	0.063 ± 0.013	0.166 ± 0.013	0.038 ± 0.009
CatBoost	$n_{\text{est}} = 2000$	0.687 ± 0.047	0.975 ± 0.011	0.063 ± 0.012	0.166 ± 0.013	0.039 ± 0.009
LightGBM	$n_{\text{est}} = 100$	0.708 ± 0.047	0.975 ± 0.013	0.069 ± 0.014	0.174 ± 0.014	0.035 ± 0.007
LightGBM	$n_{\text{est}} = 200$	0.706 ± 0.061	0.974 ± 0.013	0.058 ± 0.011	0.159 ± 0.013	0.041 ± 0.008
LightGBM	$n_{\text{est}} = 500$	0.644 ± 0.050	0.972 ± 0.015	0.055 ± 0.011	0.154 ± 0.013	0.049 ± 0.010
LightGBM	$n_{\text{est}} = 1000$	0.604 ± 0.055	0.971 ± 0.015	0.054 ± 0.011	0.153 ± 0.013	0.050 ± 0.011
LightGBM	$n_{\text{est}} = 1500$	0.598 ± 0.058	0.970 ± 0.015	0.054 ± 0.012	0.153 ± 0.013	0.051 ± 0.012
LightGBM	$n_{\text{est}} = 2000$	0.596 ± 0.058	0.970 ± 0.016	0.054 ± 0.012	0.153 ± 0.014	0.051 ± 0.012
Random Forest	$n_{\text{est}} = 2000$	0.679 ± 0.080	0.976 ± 0.013	0.067 ± 0.015	0.172 ± 0.017	0.038 ± 0.007
Random Forest	$n_{\text{est}} = 1500$	0.677 ± 0.079	0.976 ± 0.013	0.067 ± 0.015	0.172 ± 0.017	0.038 ± 0.007
Random Forest	$n_{\text{est}} = 1000$	0.672 ± 0.084	0.976 ± 0.013	0.067 ± 0.015	0.172 ± 0.017	0.038 ± 0.007
Random Forest	$n_{\text{est}} = 500$	0.656 ± 0.081	0.976 ± 0.013	0.067 ± 0.015	0.171 ± 0.017	0.038 ± 0.007
Random Forest	$n_{\text{est}} = 200$	0.630 ± 0.073	0.976 ± 0.013	0.066 ± 0.015	0.170 ± 0.018	0.038 ± 0.007
Random Forest	$n_{\text{est}} = 100$	0.593 ± 0.095	0.976 ± 0.013	0.064 ± 0.015	0.168 ± 0.017	0.038 ± 0.008
XGBoost	$n_{\text{est}} = 200$	0.706 ± 0.051	0.973 ± 0.014	0.062 ± 0.015	0.164 ± 0.017	0.038 ± 0.009
XGBoost	$n_{\text{est}} = 100$	0.692 ± 0.048	0.975 ± 0.013	0.067 ± 0.015	0.171 ± 0.016	0.036 ± 0.008
XGBoost	$n_{\text{est}} = 500$	0.668 ± 0.048	0.972 ± 0.014	0.058 ± 0.015	0.158 ± 0.017	0.041 ± 0.010
XGBoost	$n_{\text{est}} = 1000$	0.643 ± 0.046	0.970 ± 0.014	0.057 ± 0.014	0.156 ± 0.017	0.043 ± 0.011
XGBoost	$n_{\text{est}} = 1500$	0.628 ± 0.048	0.970 ± 0.015	0.056 ± 0.014	0.155 ± 0.017	0.044 ± 0.011
XGBoost	$n_{\text{est}} = 2000$	0.619 ± 0.053	0.969 ± 0.015	0.056 ± 0.014	0.154 ± 0.016	0.044 ± 0.011

4.2. Quantum Model Results

4.2.1. QSVM

The QSVM (ZZFeatureMap reps = 2, full entanglement, $C = 500$) reaches a mean AUPRC of 0.240 ± 0.109 over the 15 fits. Its structural counterpart, the RBF-kernel SVM, reaches 0.690 on the same 4-qubit PCA data with the same SVM solver, a gap of about 0.45 AUPRC in favor of the classical kernel. Since the only thing that differs between the two is the kernel, this isolates the effect of the quantum feature map: at four qubits the ZZFeatureMap Hilbert space does not capture more useful geometric structure than the Gaussian kernel for this feature distribution.

4.2.2. VQC

The best VQC (EfficientSU2, reps = 3, COBYLA, 32 parameters) reaches a mean AUPRC of 0.494 ± 0.139 , about 0.25 above the QSVM, and is better behaved at the decision threshold: precision 0.217 against the QSVM's 0.066. The QSVM, for its part, keeps a high recall (0.868) but at the cost of many false positives. The variational model's edge comes from end-to-end training: its parameters shape the decision surface to the task rather than relying on a fixed kernel geometry. Both quantum models, though, land below every classical baseline (Table 3), which we take up in the discussion.

Table 3. All models on the noise-free task, best configuration of each, ordered by average precision ($n_{\text{train}} = 656$, four-qubit PCA features, full unbalanced test fold). Values are mean $\pm$ standard deviation over 15 fits. At four qubits the classical baselines lead on AUPRC and the two quantum models fall below them.

Model	AUPRC	AUROC	Precision	Recall	F1	MCC
CatBoost	0.741 $\pm$ 0.043	0.977 $\pm$ 0.012	0.046 $\pm$ 0.008	0.901 $\pm$ 0.044	0.088 $\pm$ 0.015	0.199 $\pm$ 0.017
SVM (linear)	0.720 $\pm$ 0.019	0.971 $\pm$ 0.016	0.163 $\pm$ 0.182	0.862 $\pm$ 0.043	0.245 $\pm$ 0.168	0.345 $\pm$ 0.140
LightGBM	0.708 $\pm$ 0.047	0.975 $\pm$ 0.013	0.036 $\pm$ 0.008	0.903 $\pm$ 0.050	0.069 $\pm$ 0.014	0.174 $\pm$ 0.014
XGBoost	0.706 $\pm$ 0.051	0.973 $\pm$ 0.014	0.032 $\pm$ 0.008	0.902 $\pm$ 0.051	0.062 $\pm$ 0.015	0.164 $\pm$ 0.017
SVM (sigmoid)	0.699 $\pm$ 0.061	0.970 $\pm$ 0.015	0.106 $\pm$ 0.107	0.864 $\pm$ 0.038	0.175 $\pm$ 0.129	0.281 $\pm$ 0.109
SVM (RBF)	0.690 $\pm$ 0.077	0.972 $\pm$ 0.014	0.101 $\pm$ 0.092	0.869 $\pm$ 0.041	0.169 $\pm$ 0.123	0.274 $\pm$ 0.103
Random Forest	0.679 $\pm$ 0.080	0.976 $\pm$ 0.013	0.035 $\pm$ 0.008	0.907 $\pm$ 0.047	0.067 $\pm$ 0.015	0.172 $\pm$ 0.017
SVM (polynomial)	0.597 $\pm$ 0.035	0.964 $\pm$ 0.023	0.354 $\pm$ 0.141	0.813 $\pm$ 0.015	0.478 $\pm$ 0.147	0.524 $\pm$ 0.118
VQC	0.494 $\pm$ 0.139	0.968 $\pm$ 0.013	0.217 $\pm$ 0.186	0.727 $\pm$ 0.053	0.305 $\pm$ 0.180	0.371 $\pm$ 0.154
QSVM	0.240 $\pm$ 0.109	0.963 $\pm$ 0.015	0.066 $\pm$ 0.019	0.868 $\pm$ 0.038	0.123 $\pm$ 0.032	0.234 $\pm$ 0.035

4.3. Consolidated Comparison (Noise-Free)

Table 3 brings all models together on the noise-free task, ordered by AUPRC. The ranking is clear, and at four qubits it does not favor the quantum models: every classical baseline, from CatBoost at 0.741 down to the polynomial SVM at 0.597, sits above the VQC (0.494) and the QSVM (0.240). The VQC takes the better precision and F1 of the quantum pair, but on the primary ranking metric the classical methods lead throughout. The most controlled contrast is the QSVM against the RBF SVM, which share everything but the kernel: the 0.45 AUPRC gap between them is the cleanest measure of what the quantum kernel costs, rather than gains, at this scale.

4.4. Noise Robustness Analysis

On imbalanced data, AUPRC is not a pure ranking statistic: at the high-recall tail its value is pulled toward the base rate π , so it can appear stable while the underlying score ordering degrades. We therefore operationalize “ranking preservation” with genuine rank statistics, Spearman ρ and Kendall τ between the noisy and noiseless decision scores (Tier 8, Table 4), and read AUPRC, AUROC and the Brier score alongside them; the curves are in Figure 8. The resulting picture is noise-stratified: the right interpretation differs by model and by noise level (a mixed hypothesis, H_{mixed}), and we use the effect-size thresholds of Cohen [48] ($\rho \geq 0.7$ strong, $\rho < 0.5$ weak) to label each regime.

Table 4. Ranking behavior of the quantum models under depolarizing noise of strength p . For each level we give AUPRC, AUROC, and the rank correlation between the noisy and noiseless decision scores (Spearman ρ , Kendall τ). Values are mean $\pm$ standard deviation over 15 fits. The rank statistics, not AUPRC, are what show whether the score ordering survives the noise.

Model	p	AUPRC	AUROC	Spearman ρ	Kendall τ
QSVM	0.001	0.240 $\pm$ 0.111	0.963 $\pm$ 0.015	0.998 $\pm$ 0.002	0.977 $\pm$ 0.015
QSVM	0.01	0.245 $\pm$ 0.109	0.966 $\pm$ 0.015	0.906 $\pm$ 0.097	0.801 $\pm$ 0.117
QSVM	0.05	0.344 $\pm$ 0.110	0.970 $\pm$ 0.017	0.520 $\pm$ 0.206	0.393 $\pm$ 0.183
VQC	0.001	0.514 $\pm$ 0.121	0.966 $\pm$ 0.016	0.723 $\pm$ 0.115	0.559 $\pm$ 0.109
VQC	0.01	0.654 $\pm$ 0.078	0.955 $\pm$ 0.016	0.371 $\pm$ 0.126	0.261 $\pm$ 0.093
VQC	0.05	0.569 $\pm$ 0.150	0.881 $\pm$ 0.036	0.006 $\pm$ 0.016	0.004 $\pm$ 0.011

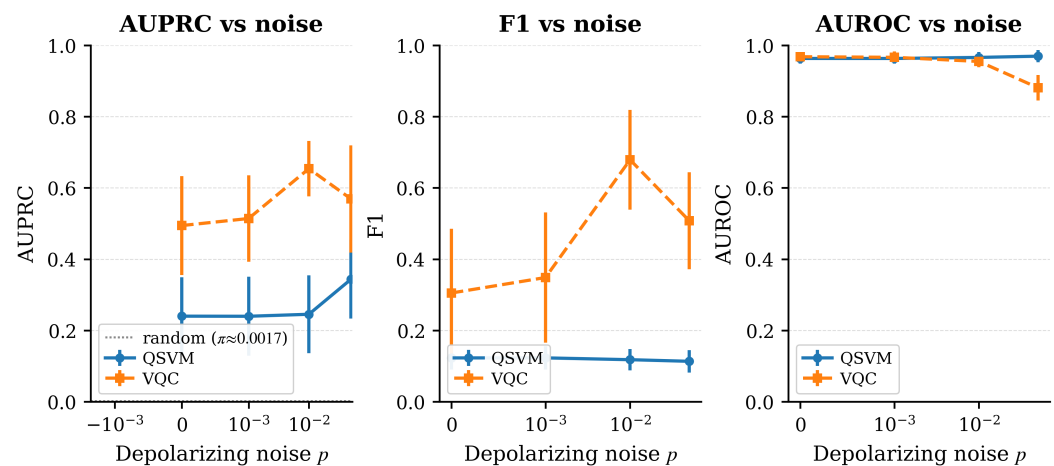


Figure 8. Quantum-model metrics against depolarizing noise of strength p : AUPRC (**left**), F1 (**centre**), and AUROC (**right**), each the mean over the 15 fits with the shaded band showing ± 1 standard deviation ($n_{\text{train}} = 656$). The noise axis is on a symmetric-log scale (linear threshold 10^{-3}) and the metric axes are pinned to $[0, 1]$. The dotted line in the AUPRC panel (**left**) marks the random-classifier level, $\text{AUPRC} = \pi \approx 0.0017$ (the fraud prevalence); both models stay well above it. The VQC AUPRC curve is non-monotonic and peaks at $p = 0.01$, while its AUROC declines steadily; that divergence is the point of this section.

For the QSVN, AUPRC is genuinely stable and even climbs somewhat ($0.240 \rightarrow 0.344$ as p goes from 0 to 0.05), AUROC holds at 0.96–0.97, and the Brier score stays in the 0.03–0.035 range throughout (decision-threshold metrics in Table 5). This stability is backed by the rank statistics, not asserted from AUPRC alone: $\rho = 0.998$ at $p = 0.001$ and 0.906 at $p = 0.01$ (both strong, H_1), relaxing to 0.520 at $p = 0.05$ (intermediate). The QSVN recall stays in the 0.87–0.89 range across all noise levels, with specificity near 0.97, so the decision threshold remains operational throughout the sweep.

Table 5. Decision-threshold behavior of the quantum models under depolarizing noise of strength p : F1, precision, recall, and the Brier score. Values are mean $\pm$ standard deviation over 15 fits. QSVN keeps recall near 0.87–0.89 throughout (it does not collapse to labeling every transaction as fraud), while the VQC Brier score rises sharply at $p = 0.05$.

Model	p	F1	Precision	Recall	Brier
QSVN	0.001	0.123 ± 0.033	0.067 ± 0.019	0.868 ± 0.037	0.035 ± 0.007
QSVN	0.01	0.118 ± 0.030	0.064 ± 0.017	0.871 ± 0.040	0.033 ± 0.007
QSVN	0.05	0.113 ± 0.032	0.061 ± 0.019	0.885 ± 0.046	0.029 ± 0.008
VQC	0.001	0.349 ± 0.183	0.261 ± 0.194	0.706 ± 0.047	0.026 ± 0.005
VQC	0.01	0.679 ± 0.140	0.749 ± 0.191	0.659 ± 0.060	0.061 ± 0.010
VQC	0.05	0.508 ± 0.136	0.762 ± 0.210	0.408 ± 0.026	0.183 ± 0.012

The VQC is where the AUPRC-versus-ranking divergence is most concrete. Its AUPRC is non-monotonic: it rises from 0.494 at $p = 0$ to a peak of 0.654 at $p = 0.01$, then dips to 0.569 at $p = 0.05$ (Figure 9), so a naive reading of AUPRC alone would report noise as beneficial. The rank statistic tells the opposite, and methodologically correct, story (Figure 10): ρ falls monotonically, $0.723 \rightarrow 0.371 \rightarrow 0.006$ (H_1 at $p = 0.001$; H_2 at $p \geq 0.01$), AUROC declines $0.968 \rightarrow 0.881$, and the Brier score worsens roughly 8-fold ($0.023 \rightarrow 0.183$), which signals a collapse of probability calibration. At $p = 0.05$ the Spearman estimate ($\rho = 0.006 \pm 0.016$, with a 95% confidence interval that crosses zero) says the noisy ordering is essentially uncorrelated with the noiseless one; the residual AUPRC of 0.569 (on a wide distribution,

$\sigma = 0.150$) reflects base-rate geometry and miscalibration, not retained discrimination. This is the precise sense in which AUPRC alone misrepresents noise behavior.

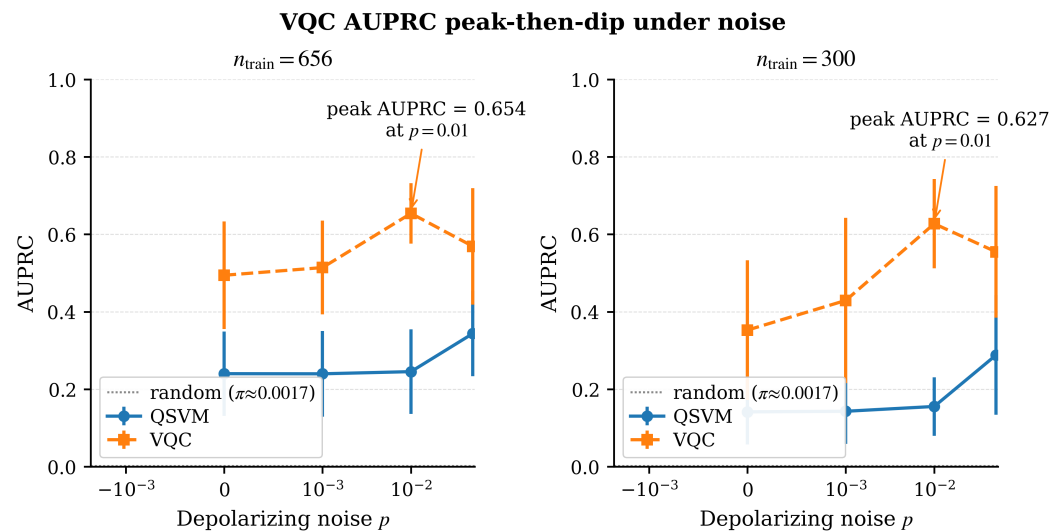


Figure 9. VQC average precision under depolarizing noise, the peak-then-dip shape in detail: 0.494 at $p = 0$, rising to a peak of 0.654 at $p = 0.01$, then easing to 0.569 at $p = 0.05$ (still far above the dotted random baseline, $\text{AUPRC} = \pi$). The left panel is the primary $n_{\text{train}} = 656$ result and the right panel the $n_{\text{train}} = 300$ sensitivity check; the shape is the same at both sizes. The QSVM curve is shown flat and low for context. Read on its own, AUPRC would suggest noise helps the VQC; the rank statistics in Table 4 show otherwise.

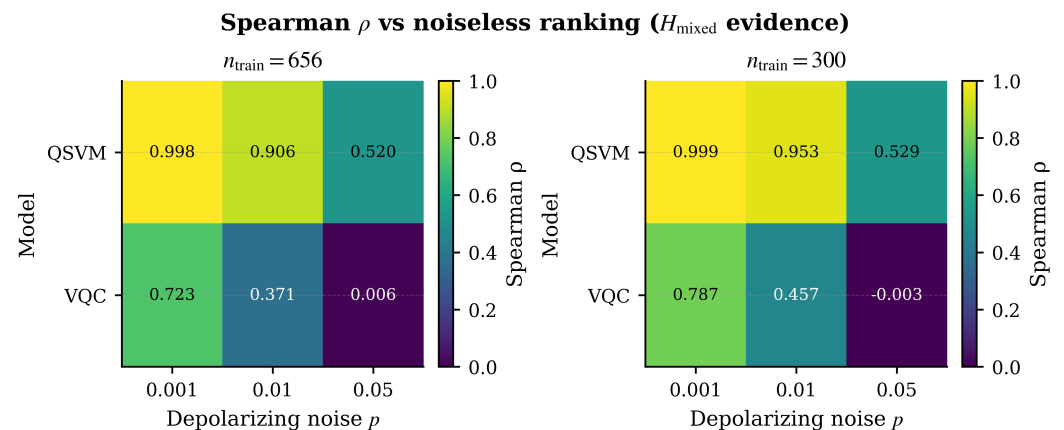


Figure 10. Spearman ρ between the noisy and noiseless decision scores, per model and noise level: the left panel the primary $n_{\text{train}} = 656$ run, the right panel the $n_{\text{train}} = 300$ sensitivity run. A higher ρ means the noise has preserved more of the score ordering. The QSVM holds $\rho \geq 0.7$ only up to $p = 0.001$ (H_1), while the VQC falls below 0.5 from $p = 0.01$ onward (H_2). The pattern is the same at both training sizes, which is why we read the noise-stratified H_{mixed} conclusion as driven by the data rather than by one design choice. The $p = 0$ column is omitted, since correlation against the noiseless reference is undefined there.

The QSVM's stronger ranking stability and the VQC's faster decay are not directly comparable: they come from qualitatively different mechanisms. The QSVM ranking inherits the convexity of the support-vector solve on a precomputed Gram matrix [44] (small kernel perturbations map to small dual-solution perturbations) and the noise resilience of the optimum under incoherent channels [26]; it has no iterative optimizer to amplify noise. The VQC, by contrast, is trained by a stochastic optimizer whose path is perturbed by both gate and sampling noise, consistent with noise-induced trainability loss [17]. As set out in Section 3.6, the two paradigms are also implemented with different best-practice

simulation strategies, so we compare them on predictive metrics while attributing their noise responses to distinct sources. Figure 11 consolidates these per-(model, noise) results across all four metrics.

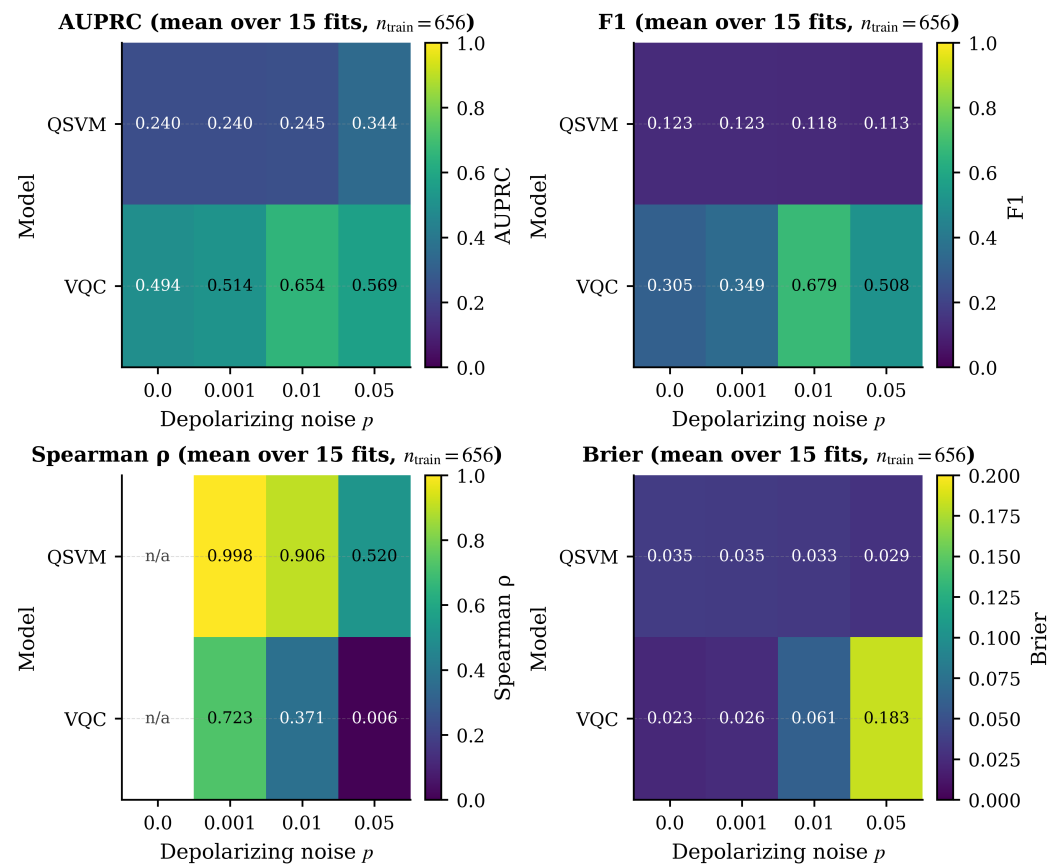


Figure 11. Per-(model, noise) heatmaps over the 15 fits at $n_{\text{train}} = 656$: AUPRC (top left), F1 (top right), Spearman ρ (bottom left), and Brier score (bottom right), each on a sequential color scale. The Spearman ρ cells at $p = 0$ carry no value, since correlation against the noiseless reference is undefined there. The Brier panel (bottom right) uses a fixed scale $v_{\text{max}} = 0.20$ to make the VQC calibration loss visible: its Brier rises from 0.023 at $p = 0$ to 0.183 at $p = 0.05$, about 8-fold. The QSVM cells in the Brier panel (bottom right) look darker, not because their Brier values are larger than the VQC's at low noise, but because the fixed scale compresses the QSVM range, which stays near 0.03 across all p .

4.5. Computational Cost Analysis

We separate two cost questions. The per-fit wall-clock times for the main experiments are reported in Section 3.6; here we look at how circuit complexity scales with the number of qubits, which is independent of the kernel implementation. Figure 12 shows circuit depth, trainable-parameter count, and total gate count across $N \in \{4, 6, 8, 10\}$. All timings were measured on a MacBook Air 15-inch (2025) with an Apple M4 chip and 24 GB unified memory, in single-thread execution; the classical models (SVM, Random Forest, XGBoost, LightGBM, CatBoost) all complete in well under a second per fit on the same hardware and serve as the timing reference.

The scaling visible in Figure 12 follows directly from the circuit structures of Section 3. The QSVM has no trainable circuit parameters; its model capacity sits entirely in the SVM dual solution. The VQC uses an EfficientSU2 ansatz of depth R_a with N qubits, which carries

$$P_{\text{VQC}}(N, R_a) = 2N(R_a + 1) \quad (5)$$

trainable rotation angles, giving $P_{\text{VQC}}(4, 3) = 32$ in agreement with the parameter count reported in Section 3. The gate count of the two paradigms scales differently in N . The ZZFeatureMap with full entanglement and R_f repetitions contains $2N$ single-qubit gates and $\binom{N}{2}$ two-qubit ZZ phases per repetition, so its total gate count grows as

$$G_{\text{enc}}(N, R_f) = R_f \left[2N + c_{\text{ZZ}} \frac{N(N-1)}{2} \right] = \mathcal{O}(R_f N^2), \quad (6)$$

where c_{ZZ} is the number of native gates produced by the chosen ZZ decomposition. The EfficientSU2 ansatz with linear entanglement carries $2N(R_a + 1)$ single-qubit rotations and $R_a(N - 1)$ CNOTs, so

$$G_{\text{ans}}(N, R_a) = 2N(R_a + 1) + R_a(N - 1) = \mathcal{O}(N R_a). \quad (7)$$

For fixed R_f and R_a , the encoding therefore scales quadratically in N while the ansatz scales linearly, which is why the QSVM circuit (an encoding-only circuit at $\text{reps} = 2$) is the deeper of the two at $N = 4$ even though it has no trainable parameters.

Two further structural costs bound the study. Density-matrix simulation requires representing a $2^N \times 2^N$ matrix, so its memory footprint is

$$\text{Mem}(N) = (2^N)^2 = 4^N \text{ complex amplitudes}, \quad (8)$$

which keeps $N \leq 10$ tractable on the M4 laptop ($\approx 10^6$ entries) but pushes $N = 16$ to $\approx 4 \times 10^9$ entries, a high-performance-computing regime rather than a desktop one. Finally, QSVM inference cost scales as

$$T_{\text{inf}}^{\text{QSVM}} = \mathcal{O}(m_{\text{train}} m_{\text{test}}) \text{ kernel evaluations}, \quad (9)$$

since each test prediction requires a fidelity evaluation against every training point, whereas a trained VQC requires only a single forward pass of $\mathcal{O}(G_{\text{enc}} + G_{\text{ans}})$ gates per test sample. This is the structural disadvantage of the QSVM that survives any choice of simulation backend.

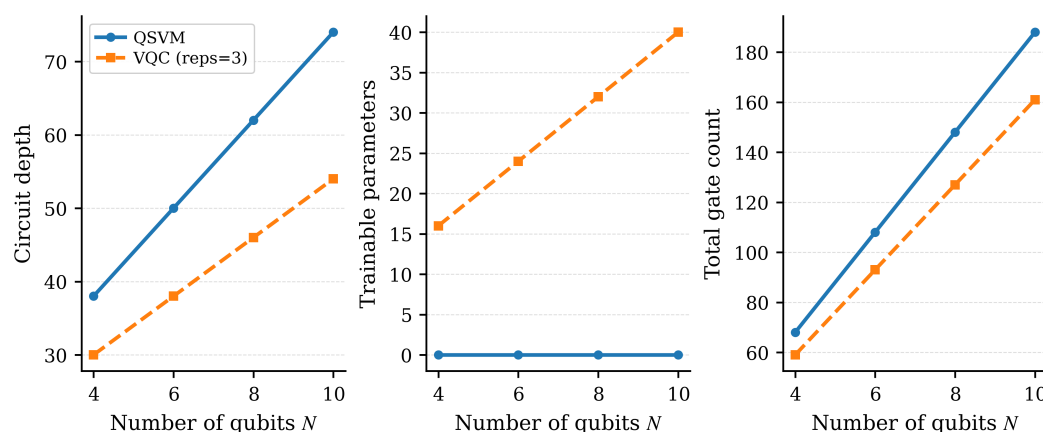


Figure 12. Quantum circuit complexity against qubit count. (Left): circuit depth. (Centre): number of trainable parameters (the QSVM has none). (Right): total gate count. At $N = 4$ the QSVM circuit has a greater depth (38) and more gates (68) than the VQC at $\text{reps} = 3$ (depth 30, 59 gates).

A counterintuitive point stands out: at the same qubit count the QSVM circuit is deeper and carries more gates than the VQC. At $N = 4$ the QSVM circuit has a depth of 38 and 68 gates, against the VQC's depth of 30 and 59 gates ($\text{reps} = 3$). One might expect the deeper circuit to be the more noise-fragile of the two, yet Section 4.4 shows the opposite for

ranking: the QSVM's kernel-method structure keeps its score ordering stable even though its circuit carries more gate error, while the VQC's ranking is the one that decays. Circuit depth therefore bounds the absolute gate-error budget, but on its own it does not predict which model's ranking survives noise. Classical simulation is in turn bounded by the 2^n growth of the density matrix, which keeps four to ten qubits tractable on the M4 laptop but turns much larger registers into a high-performance-computing problem; this is one practical reason the study sits at four qubits on a simulator rather than at scale.

5. Discussion

5.1. Interpretation of Results

The underperformance of quantum models relative to classical baselines is consistent with the broader QML benchmarking literature [12,14,39]. At $N = 4$ qubits, the quantum feature space has dimension $2^4 = 16$, comparable to a simple polynomial kernel, which is insufficient to provide meaningful expressivity advantages over well-tuned classical kernels operating on a feature space that is already PCA-conditioned. This aligns with the theoretical analysis by Liu et al. [13], who show that quantum advantage in kernel methods requires exponentially hard-to-classically-simulate problem structure that is absent in standard tabular datasets.

Among the quantum pair, the VQC clearly leads the QSVM: AUPRC 0.494 against 0.240, with a better-calibrated decision boundary (precision 0.217 against 0.066). The end-to-end training of the VQC lets it adapt its decision surface to the task, which the fixed ZZFeatureMap kernel cannot do. We are careful not to read this as a quantum advantage, however. Both quantum models sit below every classical baseline at four qubits, so the VQC's edge is over the QSVM, not over the classical reference.

5.2. The Noise Asymmetry and Its Implications

The noise behavior splits cleanly by model, and not quite in the direction one might expect. The QSVM is the more robust of the two at realistic noise: its score ranking is essentially intact through $p = 0.01$ (Spearman $\rho = 0.998$ then 0.906), its calibration holds (Brier near 0.03), and its recall stays around 0.87–0.89 across the sweep, so the decision threshold remains operational. The VQC is the opposite case: its AUPRC is actually highest at moderate noise (the peak at $p = 0.01$), yet its ranking decays monotonically (ρ from 0.72 down to essentially zero) and its calibration worsens roughly 8-fold. The lesson is that AUPRC on its own can move in the wrong direction relative to the underlying ranking, which is the methodological point of the noise sweep.

For near-term deployment the picture is therefore nuanced rather than a clean winner. In a low-noise regime ($p \lesssim 10^{-3}$, within reach of the best current hardware [27]), the QSVM's ranking stability is appealing; at higher noise its ranking weakens but the decision threshold remains operational. The VQC keeps a usable decision threshold across the range, but its AUPRC should not be read alone, since noise flatters that one metric. Neither model beats the classical baselines at four qubits, so at this scale the practical recommendation stays the classical one.

5.3. Computational Considerations

At the full evaluation scale ($n_{\text{train}} = 656$, the complete test fold), a single QSVM fit takes about 3.8 min with the statevector kernel at $p = 0$ and about 25 min with the density-matrix kernel at $p > 0$, while a VQC fit takes about 14 and 10 min respectively (Section 3.6); the classical baselines finish in well under a second. The kernel pivot (statevector for $p = 0$, density matrix for $p > 0$) is what made the full-fold evaluation feasible at all, since the naive circuit-pair kernel did not scale to $n_{\text{test}} \approx 95,000$. One structural caveat survives regardless

of implementation: QSVM inference cost grows with the training-set size, because each prediction needs a kernel evaluation against every training point, whereas a trained VQC needs only a single forward pass.

5.4. Comparative Analysis

Set against prior quantum machine-learning work on credit-card fraud, the contribution of this study is methodological rather than performance-based. At four qubits, the QSVM and VQC here reach AUPRC values of 0.240 and 0.494, well below the tuned classical band of 0.68 to 0.74, so no quantum advantage on this task is claimed. The closest prior study, El Alami et al. [40], runs VQC and QNN variants on the same ULB dataset but balances the classes before training and reports a single best $F_1 = 0.88$; the imbalance-preserved AUPRC reported here is a stricter operating point and a closer match to deployment conditions. Chaves et al. [41] report average precision 0.793 ± 0.085 for an autoencoder-VQC mixture of experts on a comparable imbalanced dataset, but the XGBoost component of that pipeline already reaches 0.770 ± 0.096 , so the quantum component contributes only when embedded in a feature-engineered hybrid. Kyriienko and Magnusson [38] document a 15% average-precision separation in favor of a re-uploading quantum kernel, but only at 20 qubits, which is consistent with the negative result at four qubits reported here.

The dataset scale and distribution sit at the upper end of this small literature: the full ULB set of 284,807 transactions is used and the native 0.172% fraud prevalence is preserved, whereas several prior studies balance the classes or use smaller subsets. The most distinctive aspect is the noise characterization. Depolarizing noise $p \in \{0, 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$ is swept on both QSVM and VQC, and each AUPRC value is paired with Spearman ρ , Kendall τ , AUROC, and the Brier score. Several prior fraud-QML papers note noise robustness without reporting per-noise-level metrics; the analysis in Section 4.4 quantifies the behavior and shows that AUPRC alone can give the wrong impression of how noise affects a model, the VQC AUPRC peaking at $p = 0.01$ while its score ranking decays monotonically. The rank-versus-AUPRC distinction is what this work adds to the literature.

5.5. Limitations

A few limitations should temper how the results are read. The sampling design (3 stratified folds $\times$ 5 seeds, 15 fits per condition) matches the repeated-cross-validation budget of Chaves et al. [41] and is well above common single-split practice, but with only ≈ 164 fraudulent cases per test fold, the bootstrap confidence intervals are necessarily wide; that dispersion is reported plainly, including outliers (one VQC fold at $p = 0.05$ gives an isolated AUPRC = 0.033, and the Spearman interval at $p = 0.05$ crosses zero). The dimensional reduction from thirty to four principal components fits the four-qubit register but is one defensible choice among many; a different dimensionality or encoding could shift the quantum–classical gap. On the noise side, a single depolarizing channel is simulated at four qubits and the results are not validated on quantum hardware; at this scale the asymptotic exponential kernel concentration of Thanasilp et al. [15] is not yet active, though its noise-induced component (their Corollary 2) does contribute to the QSVM kernel flattening observed here. The VQC uses COBYLA with $\text{max_iter} = 500$; a different optimizer or budget could improve trainability and, in turn, the measured noise robustness, so the VQC numbers should be read as a lower bound in that sense. Following Bowles et al. [39], the absolute values are also subject to a benchmark lottery across datasets and splits, and the Cohen [48] effect-size thresholds ($\rho \geq 0.7$, < 0.5) are used here only as interpretive labels, not sharp decision rules.

The noise model itself is deliberately simple. AerSimulator applies independent per-gate depolarizing errors, whereas real hardware adds correlated errors, crosstalk, readout

error, and T_1/T_2 relaxation [27]; no error mitigation [16,49] is applied, so the figures here describe a worst-case NISQ scenario rather than a tuned one. The study is also confined to four qubits and a single dataset; a different scale or problem structure could change the comparison.

5.6. Future Directions

Natural extensions include: evaluation on real IBM Quantum hardware to validate the simulation findings under realistic noise conditions; application of error mitigation techniques [16,49] to assess their recovery potential for QSVM and VQC; exploration of larger qubit counts ($N = 16$ to 20) with HPC-accelerated simulation; investigation of noise-aware training [26] where VQC parameters are optimized under the target noise level; and benchmarking against attention-based deep learning architectures, which have shown strong results on related multi-step sequential prediction tasks [50], as an additional classical reference for transaction-sequence variants of the problem. Two complementary quantum directions are hybrid transfer-learning designs that combine classical and quantum networks [51], and analyses of the expressive power and effective dimension of the quantum models themselves [52], both of which could clarify when the quantum components contribute beyond the fixed four-qubit architectures studied here.

6. Conclusions

This paper presented an empirical comparison of quantum kernel methods (QSVM) and variational quantum classifiers (VQCs) against classical baselines (SVM, Random Forest, XGBoost, LightGBM, CatBoost) for credit card fraud detection, together with a noise robustness analysis and a computational cost characterization. The main conclusions are:

1. At $N = 4$ qubits, the classical baselines (AUPRC 0.60–0.74) outperform both quantum models (VQC 0.494, QSVM 0.240). The controlled QSVM-versus-RBF-SVM comparison, which isolates the quantum kernel, shows a gap of about 0.45 AUPRC in favor of the classical Gaussian kernel, so at this scale the quantum feature map does not help.
2. Noise-free, the VQC is the stronger of the two quantum models (AUPRC 0.494 vs. 0.240, precision 0.217 vs. 0.066); its end-to-end training yields a better-calibrated boundary than the fixed ZZFeatureMap kernel. Both, however, stay below every classical baseline.
3. Under depolarizing noise the two models differ in kind, not just degree. The QSVM keeps its score ranking (Spearman $\rho = 0.998$ at $p = 0.001$, 0.906 at $p = 0.01$) and a stable, operational decision threshold (recall 0.87–0.89). The VQC's AUPRC is non-monotonic (a peak at $p = 0.01$), while its ranking decays monotonically to near zero, so AUPRC alone misrepresents its noise behavior. Neither model beats the classical reference at four qubits.
4. The kernel pivot (statevector fidelity at $p = 0$, density matrix at $p > 0$) made the full-fold evaluation tractable; per fit the QSVM costs about 3.8 min noise-free and 25 min under noise, the VQC about 14 and 10 min, and the classical models under a second. QSVM inference also scales with the training-set size, a structural disadvantage independent of the simulator.

The broader message is methodological: under noise, AUPRC has to be read together with a genuine rank statistic such as Spearman ρ , because noise can leave AUPRC flat, or even raise it, while the underlying ranking quietly degrades. For four-qubit fraud detection on this dataset, the honest conclusion is that the classical baselines remain the better practical choice, and the contribution of the quantum study is in the noise-robustness methodology rather than a performance advantage.

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